Aviral

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PROFESSIONAL SUMMARY

Machine Learning & Software Engineer specializing in Computer Vision, Physics-Informed ML, and High-Performance Backend Systems. Proven track record architecting multi-mode aerial surveillance pipelines (Siamese Networks, YOLOv8-OBB, ByteTrack) and astrodynamic collision forecasting systems aligned with NASA CARA research guidelines. Proficient in PyTorch, FastAPI, React/TypeScript, and scalable spatial intelligence full-stack deployments.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, SQL, HTML5, CSS3
AI & Deep Learning: PyTorch, Ultralytics YOLOv8 (Standard & OBB), XGBoost, Siamese Networks, ByteTrack, Scikit-Learn
Vision & Astrodynamics: OpenCV, SSIM, Remote Sensing (xBD, xView2, DOTA v1.5, VisDrone), SGP4 Orbital Mechanics
GenAI & Explainability: SHAP (Feature Attribution), Retrieval-Augmented Generation (RAG), Mistral LLM, Operational Triage
Backend, Web & Cloud: FastAPI, React, Vite, Three.js, REST APIs, SQLite, MongoDB, Azure, Docker, Git

PROJECTS

AERION — AI Satellite & Drone Geospatial Intelligence Platform

Live Demo: aerion.aviral17.me | GitHub: AVIRAL017/AERION | Technologies: PyTorch, YOLOv8-OBB, OpenCV, FastAPI, Azure, Mistral AI

- Engineered an end-to-end aerial intelligence platform with dual operational modes: post-disaster building structural damage assessment and autonomous border/perimeter tactical surveillance.
- Trained a Siamese ResNet-18 on benchmark xBD and xView2 datasets, combining hybrid feature-difference maps with SSIM to classify damage across 4 severity tiers (No Damage, Minor, Major, Destroyed).
- Fine-tuned YOLOv8s on VisDrone for small aerial targets and YOLOv8n-OBB on DOTA v1.5 for arbitrary-angle orientation without overlap ambiguity; integrated ByteTrack for persistent multi-object tracking across flight paths.
- Architected a 3-tier operational triage engine (CRITICAL, HIGH, MEDIUM) preserving raw coordinate provenance, coupled with Mistral RAG synthesis to generate structured tactical intelligence briefs.

Perigee — Explainable AI Satellite Conjunction & Collision Assessment

GitHub: AVIRAL017/Perigee | Technologies: Python, SGP4 Astrodynamics, XGBoost, SHAP, Mistral AI, Three.js, FastAPI

- Developed an explainable astrodynamics AI platform forecasting Low Earth Orbit (LEO) satellite collision risks, directly aligned with research priorities established by NASA's Conjunction Assessment Risk Analysis (CARA) initiative.
- Constructed a self-throttled ingestion pipeline caching live Conjunction Data Messages (CDMs) and TLEs from Space-Track.org into SQLite with rate-limit exponential backoff, preventing service interruptions.
- Propagated 3D orbital state vectors via SGP4 mechanics to extract 9 astrodynamic features (closing velocity, miss distance, covariance scaling, radar cross-section, radial/in-track/cross-track deltas ΔU, ΔV, ΔW).
- Tuned an XGBoost classifier achieving 79% accuracy across hazard tiers; implemented SHAP feature attributions, automated Mistral maneuver briefings, and an interactive Three.js 3D orbital trajectory viewer.

RAKSHAK — Predictive Safety & Emergency Dispatch Platform

GitHub: AVIRAL017/RAKSHAK | Technologies: Python, Flask, Scikit-Learn, Gradient Boosting, Open-Meteo, Leaflet, REST APIs

- Developed a predictive safety platform leveraging dual Gradient Boosting classifiers to forecast accident probability and severity hotspots across 149 Indian cities.
- Integrated real-time weather, traffic density, and regional seismic telemetry into a multi-factor sensor fusion engine to drive intelligent, proactive ambulance pre-positioning and minimize first-responder transit latency.

CERTIFICATIONS & ACHIEVEMENTS

- **ISRO Certification:** Participated in ISRO Hackathon.
- **Coursera Specializations (DeepLearning.AI):** Verified in Neural Networks & Deep Learning, Advanced Learning Algorithms, and Generative AI.
- **Hack with Uttar Pradesh 2025 (Chandigarh University):** State-level competitive hackathon participant.
- **HackerRank Problem Solving:** 4-Star Coder in Problem Solving & Algorithms (Profile: AIML1B_059).

EDUCATION

Pranveer Singh Institute of Technology (PSIT) — Kanpur, India
Bachelor of Technology in Computer Science & Engineering (AI & ML) | CGPA: 8.034 / 10.0 | 2024 – 2028

Mount Litera Zee School — Etawah, India
Higher Secondary Certificate (Class XII - PCM) | Aggregate: 80.0% | 2024